Tutorial 3.1
Constructor and Destructor

Overview

- This tutorial covers topic about how to define constructor and destructor methods.
- You will be doing this tutorial in a collaborative coding session in groups.

Breakout Sessions

- I will split the main meeting room on Webex into several rooms. Each room will have 4 members.
- You will be assigned to a room. See the attached file to find which room you belong to.

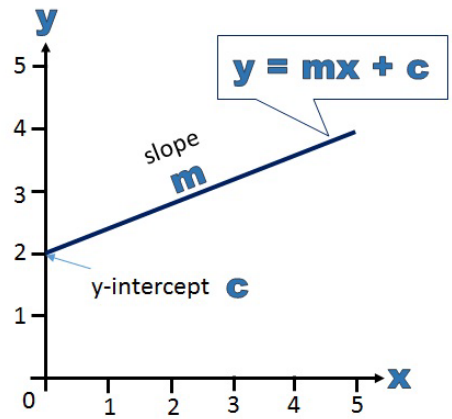
Collaborative Coding

- In each room, appoint one member to be the host. The host member will initiate a session for collaborative coding using Live Share.
- Other members will be invited as collaborators.
- All members (host and collaborators) will need to open VS Code.
- The host member will also need to share his/her screen via Webex.
- Each group is required to present the code at the end of this session.

Problem

In this tutorial, you will define a class named `Line` to model a straight line equation. Each line can be expressed by the equation, **y = mx + c**

where *m* is the slope of the line and *c* is the y-intercept, i.e. the y coordinate of the location where the line crosses the *y*-axis (see the following figure)



Complete the program **tutorial.cpp** according to the following tasks:

1. Declare and define the class `Line`. You can use either inline or separation style for these tasks. Note that, the attributes for the class have been given in the class declaration.

   Add the following methods to the class:

   a. A default constructor which sets all the attributes to zero values.
   b. A constructor that accepts two parameters to set the attributes *m* and *c,* respectively.
   c. A constructor that accepts four parameters which are **`x1`**, **`y1`**, **`x2`**, and **`y2`**, representing the coordinates x and y of two points that the line passes through. The slope and y-intercept of the line are then calculated as follows:

$$m = \frac{y2 - y1}{x2 - x1}$$

$$c = y1 - mx1$$

   d. A destructor which shows a message, for example **"The object is being destroyed"**.
   e. Accessor (or getter) methods for all the attributes,.i.e. `getM()` and `getC()`
   f. Mutator (or setter) methods for all the attributes.
   g. A method named `printEquation()` that print the equation of the straight line. There are three types depending on the value of the y-intercept, *c*:

- `Eq.  y=mx` *(if the y-intercept is zero)*
- `Eq.  y=mx-c` *(if the y-intercept is negative)*
- `Eq.  y=mx+c` *(if the y-intercept is positive)*

2. In the function main(),
   a. create a `Line` object named `line1` using an appropriate constructor to model the line: **y=2x-5**
   b. and print the equation onto the screen using an appropriate method.

3. Create another `Line` object named `line2` using an appropriate constructor to model a line that passes through these two points: **(0,0)** and **(2,8)**, and print the equation onto the screen.

4. Another `Line` object named `line3` has been created using the default constructor (already given in the code). Complete this task such that the slope, *m* and y-intercept, *c* of the line are specified with the values entered from the keyboard, and print the equation onto the screen.

Output:

Expected result of the program is as shown in the following figure. **Bold** text indicates keyboard inputs.

```
Eq.  y=2x-5

Eq.  y=4x

Enter the slope and y-intercept of the line,  m  c => 3 10

Eq.  y=3x+10
```